

Problem 1: Using filter and difference equation

Matlab provides the filter function which is a Direct Form-II transposed implementation of the standard difference equation. Such functions save coding time and help us to focus more on the system behavior.

- Re-do problem no. 3 from our classwork-1's handout and implement $y(n) = \frac{x(n)+x(n-1)+x(n-2)}{3}$ using the **'filter'** command to filter out the high frequency components of the ECG signal.
- Instead of using 'filter' command, convolute the ECG signal with $h = [1/3, 1/3, 1/3]$ using **'conv'** function. Compare the result with that of (a).
- Write your own convolution function and re-do (b) using your convolution function.

Problem 2: Basics about sinusoids and sampling rate

Let the sampling rate f_s be 50 Hz, and the time array t varies from 0 to 1 second.

- Create and plot a sine signal $y_1(t) = \sin(2\pi f_1 t)$, where $f_1 = 10 \text{ Hz}$
- Create and plot a sine signal $y_2(t) = \sin(2\pi f_2 t)$, where $f_2 = 60 \text{ Hz}$
- Can you differentiate between $y_1(t)$ and $y_2(t)$ plot? Why?

Problem 3: Signal data to sound

Let the sampling rate f_s be 50 kHz, and the time t varies from 0 to 1 second.

- Create and plot a 2 kHz signal $y_1(t) = \sin(2\pi f_1 t)$ using Matlab. Let the horizontal axis vary from 0 to 0.01 second. Use the command **'sound(signal, sampling rate)'** to listen the tone.
- Create, plot and listen to the sine signal $y_2(t) = \sin(2\pi f_2 t)$, where $f_2 = 6 \text{ kHz}$. How different is the sound of y_2 in comparison to the sound of y_1 ?
- Create, plot and listen to the sine signal $y_3(t) = \sin(2\pi f_3 t)$, where $f_3 = 25 \text{ kHz}$. Can you still hear the sound?